The deletion transform, reduced

Exact Uvarov algebra for occupied Fejér, the τ-quotient, and where cancellation actually sits

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Abstract

Round 390 identified occupied Fejér as node deletion from a discrete Bernstein–Szegő measure (full cosine grid, $\gamma_k = \frac{1}{4}$). Round 389 placed the remaining cancellation load in the μ-OP Chebyshev Gram. This note derives the closed Uvarov formula. What is proved: if σ is finite atomic and Ξ a subset of its support with masses W, writing $\tau_n = \det(I - K_n[\Xi]W)$ for the Christoffel–Darboux Gram of σ at the deleted nodes ($\tau_0 = 1$), the occupied monic coefficients are

$$\gamma_n(\mu) = \gamma_n(\sigma) \cdot \frac{\tau_{n+1}\,\tau_{n-1}}{\tau_n^2}.$$

One-point and block forms agree over $\mathbb{Q}$; iterated Sherman–Morrison is the stable evaluation. On FRAME-A $w = 9$ the identity holds at $9 \cdot 10^{-15}$ against occupied Fejér. Bernstein–Szegő specialises to $\gamma_n(\mu) = \frac{1}{4} \cdot \tau_{n+1}\tau_{n-1}/\tau_n^2$ up to the r390 residual $1.85 \cdot 10^{-6}$. Assist 0.0399 and $\mathrm{d}\Delta_{12} = 0.04946$ are exact Uvarov readouts of K^μ. What is *not* proved: F_ε from $\rho_{\mathrm{AP}} < 1/5$. Two-period AP deletion ($\rho_{\mathrm{AP}} = 1$) stays at $\gamma = \frac{1}{4}$; a consecutive ν-run of length 3 leaves the box; the kz 55 jump 0.3942 against $\frac{2}{5}$ is census. Cancellation of Assist lives in the *signs* of $K^\mu[\Xi]$, not in the positive τ-quotients of γ. No RH claim.

Status. Lemmas 1–2 are proved (finite identities). Lemma 3 is a machine identity. Proposition 1 is a finite census, not a cofinal law. Proposition 2 names the reduction. No RH claim. No L^* claim. No G_ε claim.

1 Recollection

Occupied Fejér of [1] is the restriction of the full cosine-grid measure $\sigma = \sum_k (1 - x_k)\,\delta_{x_k}$, $x_k = \cos(2\pi k/L)$, $L = 4h - 2$, to the μ-nodes. The complement Ξ is the ν-support. The full grid is discrete Bernstein–Szegő: $\max_{k\ge 2}|\gamma_k(\sigma) - \frac{1}{4}| = 1.84 \cdot 10^{-6}$ on FRAME-A $w = 9$. G^μ_ε reduced to the pair $(F_\varepsilon, W_\varepsilon)$: Fejér-on-occupied, and the r342 digamma/tent multiplier. This note attacks F_ε as exact transformation theory.

2 The Uvarov ratio

Let $\sigma = \sum_i w_i \delta_{x_i}$ be finite, positive, and atomic, with monic orthogonal polynomials P_n, Hankel determinants $\Delta_{n-1} = \det(m_{i+j})_{i,j=0}^{n-1}$, and $\gamma_n = \Delta_n\Delta_{n-2}/\Delta_{n-1}^2$. Let $K_n(x, y) = \sum_{j<n} \pi_j(x)\pi_j(y)$ be the orthonormal Christoffel–Darboux kernel. A Uvarov perturbation $\tilde\sigma = \sigma + \sum_{a\in\Xi} M_a \delta_{\xi_a}$ updates the Hankel by a rank-m Gram: $\tilde H_n = H_n + VMV^\top$.

Lemma 1 (Hankel determinant)**.** *Writing $\tau_n = \det(I + K_n[\Xi]M)$ with $\tau_0 = 1$,*

$$\frac{\tilde\Delta_{n-1}}{\Delta_{n-1}} = \tau_n.$$

Consequently

$$\tilde{\gamma}_n = \gamma_n \cdot \frac{\tau_{n+1}\,\tau_{n-1}}{\tau_n^2}, \qquad n \geq 1.$$

Exact over $\mathbb{Q}$ on a five-atom toy (delete last: $\tilde{\gamma} = \frac{5}{48}, \frac{8}{75}, \frac{5}{81}$; two-point indices $(1,3)$: $\tilde{\gamma} = \frac{8}{45}, \frac{4}{15}$). The block determinant and the iterated one-point form agree.

Proof. The matrix determinant lemma gives $\det(\tilde{H}_n) = \det(H_n)\det(I + MV^\top H_n^{-1}V)$. The monomial reproducing kernel is $K_n(\xi_a, \xi_b) = (V^\top H_n^{-1} V)_{ab}$, so the second factor is τ_n. The empty sum $K_0 = 0$ forces $\tau_0 = 1$. The γ-identity is the three-term ratio of Hankels. Iterating the $m = 1$ case multiplies the scalar τ-ratios; the block determinant is the same product (sequential Schur). □

Deletion of existing masses is the case $M = -W$. Then $\tau_n = \det(I - K_n[\Xi]W)$. The numerically stable evaluation on MAIN is iterated Sherman–Morrison: after deleting ξ with mass w,

$$\tau_n = 1 - w\,K_n(\xi,\xi), \qquad \tilde{K}_n(x,y) = K_n(x,y) + \frac{w}{\tau_n} K_n(x,\xi)K_n(\xi,y),$$

then proceed to the next node. On a clustered block of deletions the floating-point SM can lose digits; the Fractions block form remains exact.

Lemma 2 (Bernstein–Szegő specialisation)**.** *If $\gamma_n(\sigma) = \frac{1}{4}$ for $n \geq 1$, then*

$$\gamma_n(\mu) = \frac{1}{4} \cdot \frac{\tau_{n+1}\,\tau_{n-1}}{\tau_n^2}.$$

On the cosine-plus-Fejér grid this holds up to the r390 residual (FRAME-A $w = 9$: $1.85 \cdot 10^{-6}$). The exact identity uses $\gamma_n(\sigma)$, not $\frac{1}{4}$.

Lemma 3 (FRAME-A identity)**.** *On FRAME-A $w = 9$ ($S = 367$, $n_\nu = 104$, depth 182), iterated Uvarov of full-grid Fejér deleting the ν-nodes reproduces occupied Fejér at $\max_k |\gamma_k^{\mathrm{Uvarov}} - \gamma_k^{\mathrm{occ}}| = 9.0 \cdot 10^{-15}$. Last twelve: $|\gamma - \frac{1}{4}| = 0.03596$, jump 0.2379, inside [5].*

3 What the formula says about F_ε

Write $r_n = \tau_n/\tau_{n-1}$. Then $\gamma_n(\mu) = \gamma_n(\sigma)\, r_{n+1}/r_n$, and

$$\log \frac{\gamma_{n+1}}{\gamma_n} = \Delta^2 \log \tau + \log \frac{\gamma_{n+1}(\sigma)}{\gamma_n(\sigma)}.$$

The box $|\gamma - \frac{1}{4}| \leq \frac{1}{16}$ is $|e^{\Delta \log \tau} - 1| \leq \frac{1}{4}$ in the Bernstein–Szegő specialisation; the jump is a second difference of $\log \tau$.

Lemma 4 (jump not implied)**.** $\log \frac{5}{3} > \frac{2}{5}$. *The jump half of G_ε is not a corollary of the $|\gamma - \frac{1}{4}|$ box (r390, restated).*

Two-period AP deletion — every other cosine node, hence $\rho_{\mathrm{AP}} = 1$ — remains Bernstein–Szegő: on $h = 40$, last twelve $|\gamma - \frac{1}{4}| = 0$ and jump 0. The hoped resonance kill of F_ε does *not* fire. A consecutive deletion run of length 3 on the same grid leaves the box (jump 0.462). Run length 2 stays in. Scramble occupation seed 3 on FRAME-A leaves by jump 0.444.

V_3' of [5] fails first at a coherent last-twelve γ-scale of 2.0; scale 1.5 stays in $\mathcal{A}_{15}$. The $|\gamma - \frac{1}{4}| \leq \frac{1}{16}$ half is the WKB input. The constant $\frac{2}{5}$ is extra regularity, not forced by $\mathcal{A}_{15}$. Raising it toward $\log \frac{5}{3} \approx 0.511$ would make the jump redundant with the box and is legal for V_3'. It is not moved in this round: the kz 55 margin 0.0058 is census, not a theorem of the τ-quotient.

4 Assist and $\mathrm{d}\Delta$ as Uvarov readouts

The IIKS Gram is $E_{ij} = \sqrt{v_i v_j}\, K_n^\mu(y_i, y_j)$. Uvarov of $\mu + \nu$ deleting the ν-masses *is* μ, so K^μ is the updated kernel of $K^{\mu \cup \nu}$. On FRAME-A $w = 9$, $k = 183$: $\lambda_{\max} = 0.999830$, $\mathrm{maxdiag} = 0.9614$, $\mathrm{assist} = 0.0399$, Gershgorin assist 13.32, cancellation 0.9970. The kernel matches the B-matrix route at $1.6 \cdot 10^{-7}$.

Occupied-Fejér K^{μ_F} dressed with the actual ν-weights gives $\lambda_{\max} = 10.12$: node deletion from Bernstein–Szegő does *not* carry Assist. The 99.7% cancellation is the sign pattern of $K^\mu[\Xi]$, not the positive τ-quotients that control γ.

The r388 split $Q = Q^T + \Delta$ is the same readout: $Q_k = \sum_i v_i \pi_k^\mu(y_i)^2$ is the diagonal of the kernel increment $K_{k+1}^\mu - K_k^\mu$. FRAME-A last twelve $|\mathrm{d}\Delta| = 0.04946$, $\|\Delta\|_\infty = 0.3347$, reproducing [4].

5 Census and adversaries

Proposition 1 (core-42 census)**.** *Occupied Fejér on the* 42 *admissible MAIN windows satisfies the last-twelve box: max* $|\gamma - \frac{1}{4}| = 0.04892$ *(kz* 55*), max jump* 0.3942 *(kz* 55*), margin* 0.0058 *against* $\frac{2}{5}$*. Builder fallback: core-*42 *only.*

Named kills. Christoffel multiplication by $V(x) = \prod_{\xi \in \Xi}(x - \xi)$ (weights $w_i |V(x_i)|$ on occupied nodes) fails the identity (small-grid error 0.055). Dropping τ_{n-1} from the ratio fails (0.122). Clustered run of 3 leaves the box. Scramble occupation seed 3 leaves by jump. AP deletion does *not* leave — it is a named non-kill of F_ε, and a kill of Assist/L^* only (r387).

6 What remains

Proposition 2 (reduction)**.** *The deletion algebra is SATZ.* F_ε *reduces to a bound on* $\Delta^2 \log \tau$ *for* F_1*-bounded* ν*-runs (max run* ≤ 2*, r382, MAIN* 85/85*) together with half-filling;* $\rho_{\mathrm{AP}} < 1/5$ *does not imply the box (AP deletion is the regular case* $\gamma = \frac{1}{4}$*). The jump half at* 0.3942 *is census. Assist and* $\mathrm{d}\Delta$ *reduce to the* Sign-Schur *lemma: a sign bound on the off-diagonal of the Uvarov image* $K^\mu[\Xi]$*, equivalently the signed* μ*-CD remainder of [3] now with an exact kernel formula.* W_ε *(the r342 multiplier) is untouched and remains the weight sibling of* F_ε*. The chain*

$$G_\varepsilon^\mu \impliedby F_\varepsilon \wedge W_\varepsilon \impliedby \big(\Delta^2 \log \tau \text{ under } F_1\big) \wedge W_\varepsilon$$

stands on the positive side; Assist stays on the signed side as Sign-Schur. No m_4*. No RH claim.*

7 Machine checks

Numbered lemmas: `verify_deletion_transform.py` (same directory). Standalone: one-point and block Uvarov over $\mathbb{Q}$; rational $S = 5, 7, 9$; Chebyshev-U; jump comparison; cosine $S = 7$; $\tau_0 = 1$. Construction: $w = 9$ identity, Assist, $\mathrm{d}\Delta$, Fejér-kernel non-Assist, Christoffel and first-order mutants, AP non-kill, clustered and scramble kills. Discovery census: `deletion_transform_probe.py` (`--smoke` / full), core-42. Exit line: `DELETION TRANSFORM VERIFIED`.

Claim boundary

Finite identities for Uvarov transformations of positive atomic measures on a cosine grid, a named reduction of F_ε to a bound on $\Delta^2 \log \tau$ under F_1, and named clustered / scramble breaks. No statement about zeros of $\zeta(s)$, in either direction. No RH claim.

References

[1] S. Hamann, *The G_ε^μ-lemma, reduced*, standalone note, August 2026, `rh/problem/g_eps_mu.tex`.

[2] S. Hamann, *The Weyl energy of cancellation, reduced*, standalone note, August 2026, `rh/problem/weyl_energy.tex`.

[3] S. Hamann, *Coherence assist, reduced*, standalone note, August 2026, `rh/problem/coherence_assist.tex`.

[4] S. Hamann, *The Δ-deformation, reduced*, standalone note, August 2026, `rh/problem/delta_deformation.tex`.

[5] S. Hamann, *The V_3'-lemma, reduced*, standalone note, August 2026, `rh/problem/v3prime_proof.tex`.

[6] S. Hamann, *The pivot-band entry lemma, reduced*, standalone note, August 2026, `rh/problem/pivot_entry_lemma.tex`.